SOFTWARE REQUIREMENTS SPECIFICATION

(SRS)

Smart Complaint Management System

Project: Smart Complaint Management System

TEAM – 8

Rakshita Basappagoud Biradar - PES2UG24CS397
Rayudu Meghana - PES2UG24CS398
Reet Anand - PES2UG24CS399
Rehaan Jose Mathew - PES2UG24CS400

Revision history

Version	Date	Author	Change summary	Approval
1.0	31-08-2025	Team - 8	SRS with diagrams embedded	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

Table of Contents

1. Introduction
2. Overall description
3. External interfaces
4. System features (detailed)
5. Non-functional requirements (detailed)
6. Quality attributes & Acceptance tests
7. UML Use-Case Diagram
8. Requirements Traceability Matrix (RTM)

1. Introduction

1.1 Purpose

This document describes the requirements for the Smart Complaint Management System (SCMS). The system is intended to provide a simple web-based way for citizens to submit complaints, follow their progress and receive the final resolution.

1.2 Scope

The system will let citizens register, submit complaints, attach supporting evidence, check complaint status and give feedback after a complaint is closed. Department staff will work on complaints assigned to them, while administrators will manage users, categories, service-level rules, assignments and reports.

1.3 Users

- Citizen: creates complaints, checks their progress and submits feedback after closure.
- Department Staff: views assigned complaints, updates their status and records resolution information.
- Admin: manages users, complaints, categories, SLA rules and system reports.

2. Overall Description

2.1 Product Perspective

SCMS will work as a standalone web application. A user will access it through a modern browser. The main parts of the system are the web interface, application server and relational database. Email or SMS notifications may also be connected when required.

2.2 Major Functions

- User registration and login
- Complaint submission with evidence upload
- Creation of a unique complaint tracking ID
- Complaint tracking with status history
- Department assignment and reassignment
- Complaint resolution and closure
- Status notifications
- Reports and dashboard
- Citizen feedback

2.3 Operating Environment

The backend may be developed using Python with Flask or Django, or PHP with Laravel. The database may use MySQL or PostgreSQL. The client side will support current desktop and mobile web browsers. Communication between the client and server will use HTTPS.

3. External Interface Requirements

3.1 User Interface

The application shall provide responsive pages for citizens, department staff and administrators. Each role will have its own relevant screens, forms and dashboard options.

3.2 Software Interfaces

- A relational database will store user details, complaints, status history and audit records.
- An email/SMS service may be used for OTPs and complaint status notifications.
- File storage will be used for complaint attachments.

3.3 Communication

All client-server communication shall use HTTPS with TLS 1.2 or a higher version.

4. System Features and Functional Requirements

The following functional requirements define the main behaviour expected from SCMS.

Req ID	Requirement	Priority	Stakeholder	Acceptance Criteria	Test Case
SCMS-F-001	Citizens shall be able to create an account using their name, email or mobile number, and password.	High	Citizen	The account is created only after successful verification.	TC-Auth-01
SCMS-F-002	Registered users shall be able to log in with valid credentials.	High	Citizen / Staff / Admin	Valid credentials create an authenticated session.	TC-Auth-02
SCMS-F-003	The system shall apply role-based access for Citizen, Department Staff and Admin users.	High	Security	Each user can access only the functions allowed for their role.	TC-Auth-03
SCMS-F-004	Citizens shall be able to submit a complaint by entering its category, description and location.	High	Citizen	Required fields are checked and the complaint is saved successfully.	TC-Lodge-01
SCMS-F-005	Citizens shall be able to add supporting image or PDF files to a complaint.	Medium	Citizen	Accepted files within the configured size limit are linked to the complaint.	TC-Lodge-02
SCMS-F-006	The system shall create a unique tracking ID when a complaint is submitted.	High	Citizen / Operations	A unique tracking ID is shown after submission.	TC-Lodge-03
SCMS-F-007	Citizens shall be able to check a complaint using its tracking ID.	High	Citizen	The correct status is displayed for a valid tracking ID.	TC-Track-01
SCMS-F-008	The system shall keep a chronological record of status changes and remarks.	Medium	Citizen / Staff	Every change records the time, user and remark.	TC-Track-02
SCMS-F-009	Authorised staff and admins shall be able to search and filter complaints by ID, category, status and date.	Medium	Staff / Admin	Search results match the selected filters and the user's permissions.	TC-Track-03
SCMS-F-010	The system shall assign a complaint to the responsible department according to its category.	High	Operations	The submitted complaint is routed to the correct department.	TC-Assign-01
SCMS-F-011	An admin shall be able to move a complaint to another department or staff member.	Medium	Admin	The new ownership is updated and the reassignment is recorded.	TC-Assign-02
SCMS-F-012	Staff and admins shall be able to change complaint status and add resolution remarks.	High	Staff / Admin	Status changes follow the defined complaint workflow.	TC-Admin-01
SCMS-F-013	Staff and admins shall be able to close a resolved complaint after entering resolution details.	High	Staff / Admin	A complaint cannot be closed without resolution details.	TC-Admin-02
SCMS-F-014	The system shall notify the citizen whenever the complaint status changes.	High	Citizen	A notification is generated after a status change.	TC-Notify-01
SCMS-F-015	Admins shall be able to generate complaint reports, and citizens shall be able to give feedback after closure.	Medium	Admin / Citizen	Reports show correct data and closed complaints accept feedback.	TC-Report-01 / TC-Feedback-01

5. Non-Functional Requirements

Req ID	Requirement	Category	Priority	Measurement	Test
SCMS-NF-001	Complaint submission and status pages shall respond within 3 seconds for at least 95% of requests during normal load.	Performance	High	95th percentile response time ≤ 3 seconds.	TC-Perf-01
SCMS-NF-002	The system shall maintain at least 99.5% monthly availability, excluding planned maintenance.	Reliability	High	Monthly uptime ≥ 99.5%.	TC-Ops-01
SCMS-NF-003	The system shall support at least 10,000 registered citizens and 1,000 new complaints per day.	Scalability	Medium	The target volume is handled without breaking NF-001.	TC-Perf-02
SCMS-NF-004	The web interface shall follow WCAG 2.1 Level AA where applicable.	Usability / Accessibility	Medium	The accessibility audit passes.	TC-UX-01
SCMS-NF-005	Complaint records and audit logs shall be kept for at least five years.	Compliance	Medium	Archived records can still be retrieved.	TC-Ops-02

5.1 Security

5.1.1 Security Objectives

- Protect citizens' personal details and complaint information from unauthorised access.
- Make sure complaint records and status history cannot be changed without leaving an auditable record.

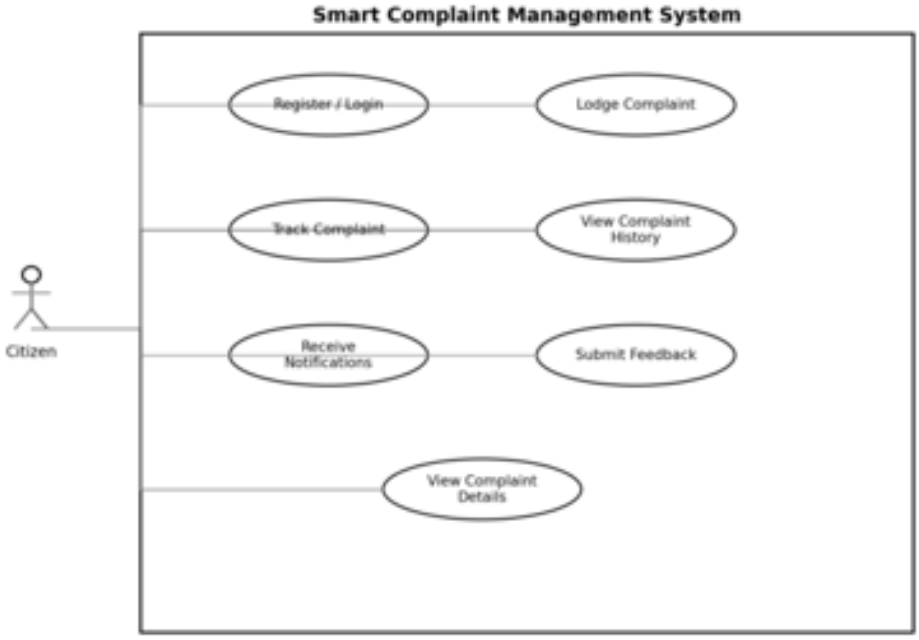
5.1.2 Security Requirements

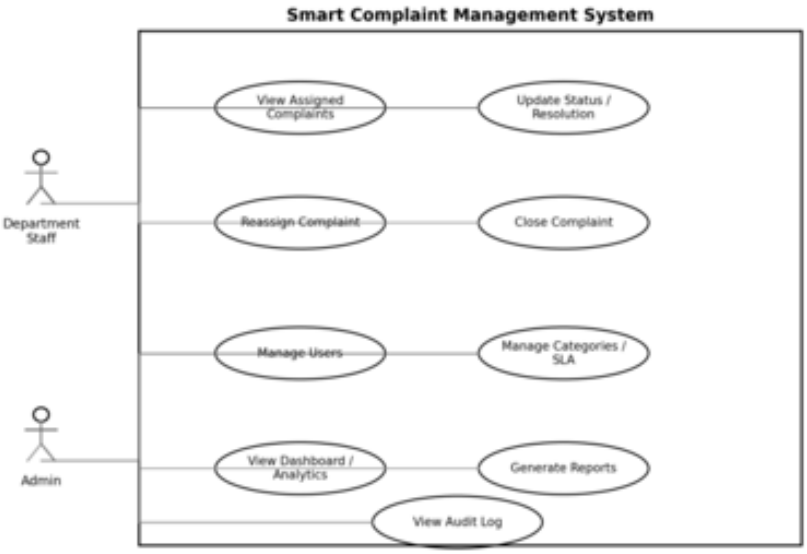
Req ID	Security Requirement	Priority	Acceptance Criteria	Test
SCMS-SR-001	All client-server communication shall use TLS 1.2 or higher.	High	No unencrypted endpoints are used.	TC-Sec-01
SCMS-SR-002	Passwords shall be stored using a salted adaptive hashing method such as bcrypt or Argon2 and shall never be kept as plain text.	High	The database and logs contain no plain-text passwords.	TC-Sec-02
SCMS-SR-003	Role-based access control shall be checked on the server for every protected function or API.	High	Unauthorised roles cannot access restricted functions.	TC-Sec-03
SCMS-SR-004	User sessions shall expire after 15 minutes of inactivity and shall use Secure, HttpOnly and SameSite cookies.	Medium	Session expiry and cookie settings are verified.	TC-Sec-04
SCMS-SR-005	Admin and staff actions shall be stored in an audit log with a timestamp, user ID and action details.	High	Protected actions create audit entries.	TC-Sec-05

6. Quality Attributes and Acceptance Tests

- Performance: important pages should meet the specified response-time target.
- Reliability: the application should meet the stated availability target.
- Security: all five security requirements should pass their corresponding tests.
- Usability: citizens should be able to submit and track complaints without unnecessary steps.
- Acceptance: all high-priority requirements must be implemented and verified, with no critical NFR or security failures.

7. UML Use-Case Diagrams





8. Requirements Traceability Matrix (RTM)

Req ID	Requirement	Section / Design	Module	Test Case	Status
SCMS-F-001	Register account	4 / DS-Auth-01	AuthModule	TC-Auth-01	N
SCMS-F-002	User login	4 / DS-Auth-02	AuthModule	TC-Auth-02	N
SCMS-F-003	Role-based access	4 / DS-Auth-03	AuthModule	TC-Auth-03	N
SCMS-F-004	Lodge complaint	4 / DS-Complaint-01	ComplaintModule	TC-Lodge-01	N
SCMS-F-005	Attach evidence	4 / DS-Complaint-02	ComplaintModule	TC-Lodge-02	N
SCMS-F-006	Generate tracking ID	4 / DS-Complaint-03	ComplaintModule	TC-Lodge-03	N
SCMS-F-007	Track complaint	4 / DS-Track-01	TrackingModule	TC-Track-01	N
SCMS-F-008	Status history	4 / DS-Track-02	TrackingModule	TC-Track-02	N
SCMS-F-009	Search/filter	4 / DS-Track-03	TrackingModule	TC-Track-03	N
SCMS-F-010	Auto assignment	4 / DS-Route-01	RoutingModule	TC-Assign-01	N
SCMS-F-011	Reassignment	4 / DS-Route-02	RoutingModule	TC-Assign-02	N
SCMS-F-012	Update status	4 / DS-Admin-01	AdminModule	TC-Admin-01	N
SCMS-F-013	Close complaint	4 / DS-Admin-02	AdminModule	TC-Admin-02	N
SCMS-F-014	Status notification	4 / DS-Notify-01	NotificationModule	TC-Notify-01	N
SCMS-F-015	Reports / feedback	4 / DS-Report-01	Reporting / Feedback	TC-Report-01 / TC-Feedback-01	N
SCMS-NF-001	Response time	5 / DS-Perf-01	WebUI / API	TC-Perf-01	N
SCMS-NF-002	Availability	5 / DS-Reliability-01	Deployment	TC-Ops-01	N
SCMS-NF-003	Scalability	5 / DS-Perf-02	WebUI / API	TC-Perf-02	N
SCMS-NF-004	Accessibility	5 / DS-UX-01	WebUI	TC-UX-01	N
SCMS-NF-005	Data retention	5 / DS-Data-01	Database / Archive	TC-Ops-02	N
SCMS-SR-001	TLS	5.1.2 / DS-Sec-01	SecurityModule	TC-Sec-01	N
SCMS-SR-002	Password hashing	5.1.2 / DS-Sec-02	AuthModule	TC-Sec-02	N
SCMS-SR-003	RBAC	5.1.2 / DS-Sec-03	AuthModule	TC-Sec-03	N
SCMS-SR-004	Session security	5.1.2 / DS-Sec-04	AuthModule	TC-Sec-04	N
SCMS-SR-005	Audit log	5.1.2 / DS-Sec-05	AuditModule	TC-Sec-05	N